

# Signals and Systems

## Computer Assignment 4

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## Contents

|          |                                                                |           |
|----------|----------------------------------------------------------------|-----------|
| <b>1</b> | <b>Laplace Transform: RLC Circuit &amp; Bode Plot</b>          | <b>2</b>  |
| <b>2</b> | <b>Laplace Transform: Closed-Loop Control System Stability</b> | <b>4</b>  |
| <b>3</b> | <b>Z-Transform: Moving Average Filter</b>                      | <b>6</b>  |
| <b>4</b> | <b>Z-Transform: Inverse via Partial Fractions</b>              | <b>9</b>  |
| <b>5</b> | <b>Application: ECG Denoising with a Notch Filter</b>          | <b>11</b> |

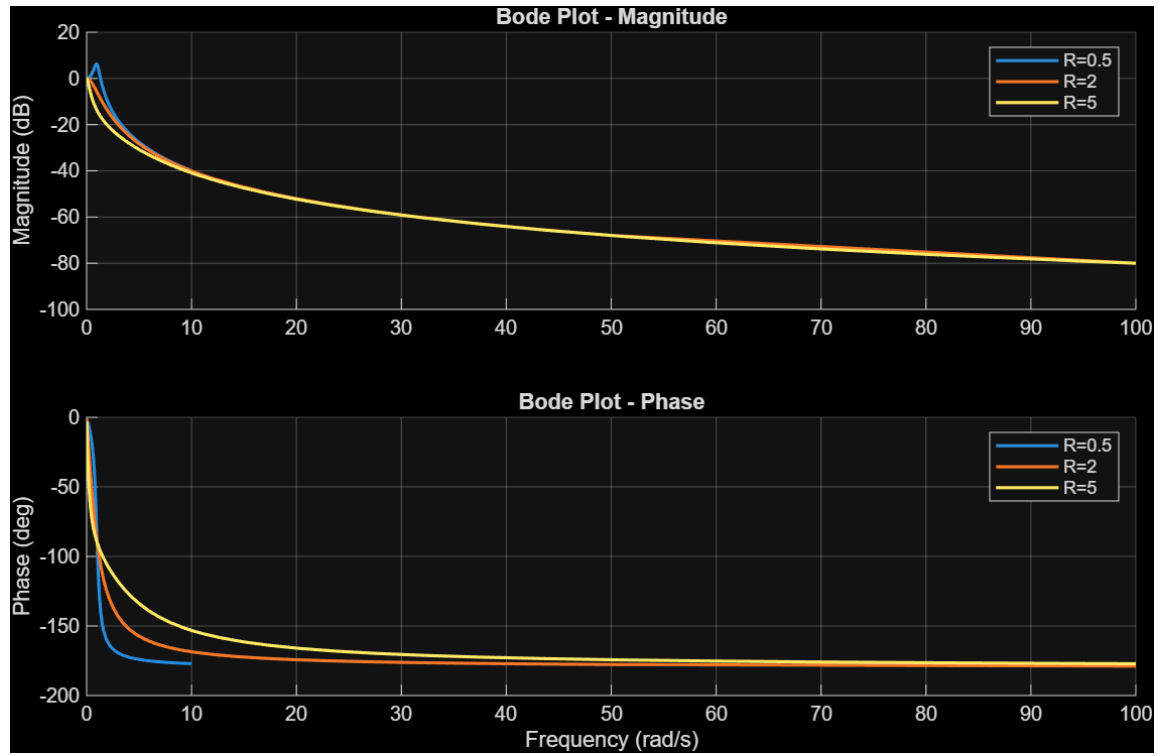
# 1. Laplace Transform: RLC Circuit & Bode Plot

(a)

Consider a series RLC circuit with input  $V_{in}$  and output  $V_C$ . Its transfer function is defined as:

$$H(s) = \frac{\omega_0^2}{s^2 + \frac{R}{L}s + \omega_0^2}, \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad (1)$$

Given  $L = 1\text{H}$  and  $C = 1\text{F}$ , the Bode diagrams for  $R = 0.5\Omega$ ,  $R = 2\Omega$ , and  $R = 5\Omega$  are plotted.



(b)

Based on the Bode plot, this system operates as a **low-pass filter**. At low frequencies (as  $\omega \rightarrow 0$ ), the magnitude approaches 0 dB (a gain of 1), meaning the signal passes without attenuation. However, at frequencies higher than the cutoff frequency, the magnitude decreases with a constant negative slope (-40 dB/decade), effectively blocking high-frequency components.

(c)

The natural frequency is calculated as  $\omega_0 = \frac{1}{\sqrt{1 \times 1}} = 1$  rad/s. This frequency represents the undamped resonance frequency of the system. In the Bode plot, particularly for low resistance values (underdamped cases), a resonance peak occurs around this specific frequency, and the phase precisely crosses  $-90^\circ$ .

(d)

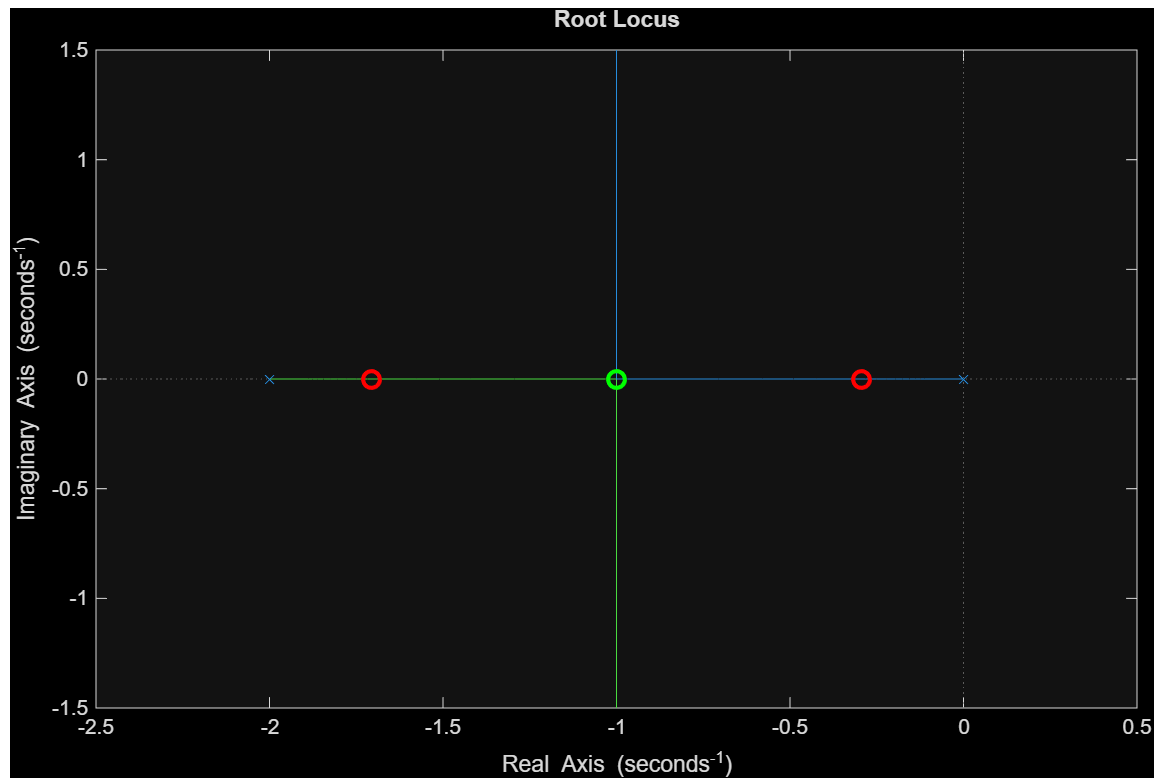
The quality factor is given by  $Q = \frac{\omega_0 L}{R}$ . For  $R = 0.5\Omega$ ,  $Q = 2$ . For  $R = 2\Omega$ ,  $Q = 0.5$ . For  $R = 5\Omega$ ,  $Q = 0.2$ .

**Relationship and Effect:** The quality factor is inversely proportional to the filter's fractional bandwidth ( $BW = \omega_0/Q$ ). Increasing the resistance  $R$  decreases  $Q$ . Consequently, a lower  $Q$  eliminates the resonance peak, widens the transition band, and flattens the frequency response near the cutoff frequency.

## 2. Laplace Transform: Closed-Loop Control System Stability

(a)

For the unity-feedback system with  $G(s) = \frac{K}{s(s+2)}$ , the closed-loop characteristic equation is  $s^2 + 2s + K = 0$ .



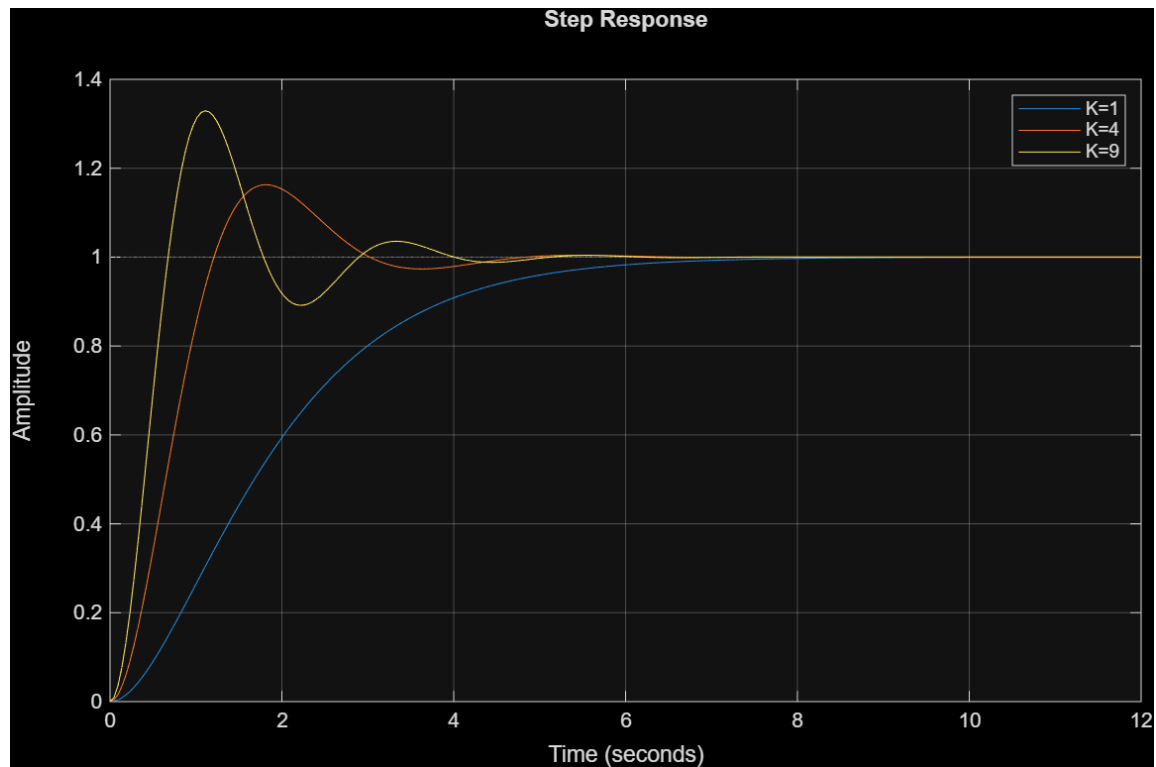
(b)

The roots of the characteristic equation are  $s = -1 \pm \sqrt{1 - K}$ . Since the real part of all closed-loop poles is strictly negative ( $-1$ ) for all  $K > 0$ , the system is always **stable**.

- $K = 0.5$ : Poles at  $s = -0.293, -1.707$ . The system is **overdamped** (real, distinct poles).

- $K = 1$ : Poles at  $s = -1, -1$ . The system is **critically damped** (real, repeated poles).
- $K = 4$  and  $K = 9$ : Poles at  $s = -1 \pm j\sqrt{3}$  and  $s = -1 \pm j\sqrt{8}$  respectively. The system is **underdamped** (complex conjugate poles with a negative real part).

(c)



**Effect of increasing  $K$ :** As  $K$  increases, the imaginary part of the poles increases, which decreases the damping ratio ( $\zeta$ ). This results in a significantly higher **overshoot** in the step response. However, the **settling time** remains largely unaffected because it is primarily determined by the real part of the dominant poles ( $T_s \approx 4/|\sigma|$ ), which remains constant at  $-1$  for  $K \geq 1$ .

### 3. Z-Transform: Moving Average Filter

(a)

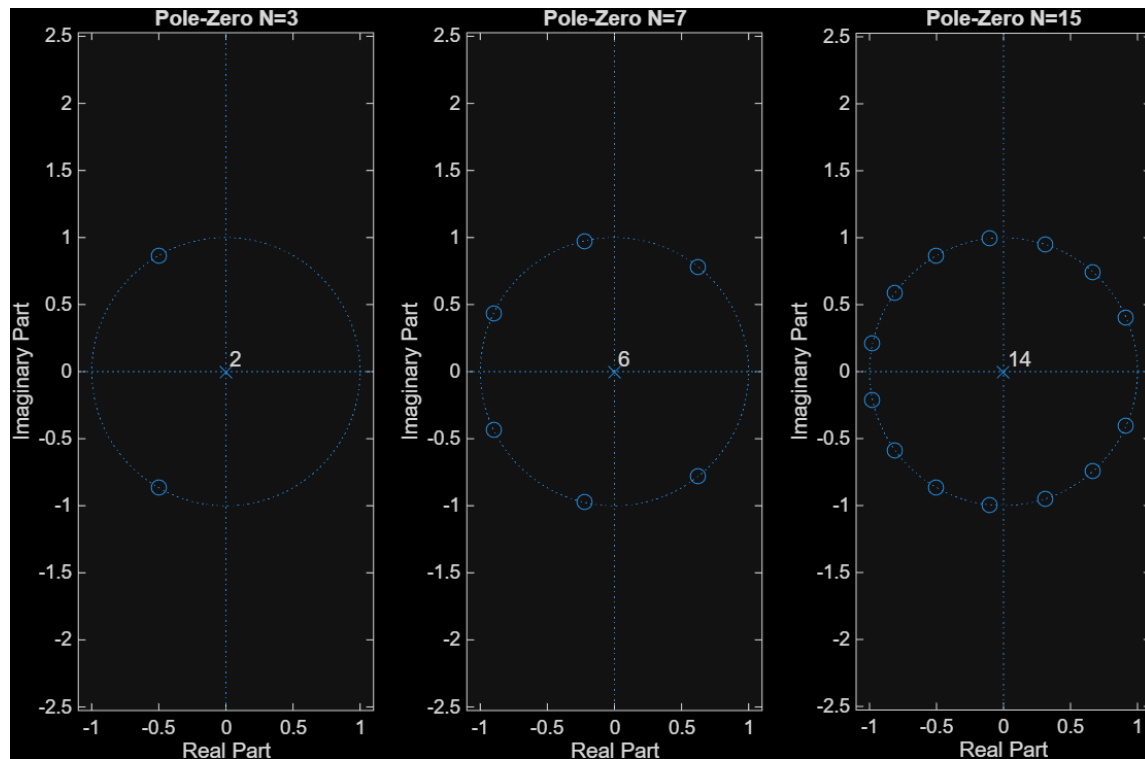
Taking the Z-transform of the N-point moving average difference equation yields:

$$H(z) = \frac{1}{N} \sum_{k=0}^{N-1} z^{-k} = \frac{1 - z^{-N}}{N(1 - z^{-1})} \quad (2)$$

This is an **FIR (Finite Impulse Response) filter**. The impulse response has a finite duration of  $N$  samples, and any pole at  $z = 1$  is explicitly canceled by a corresponding zero at  $z = 1$ , meaning there is no infinite feedback loop.

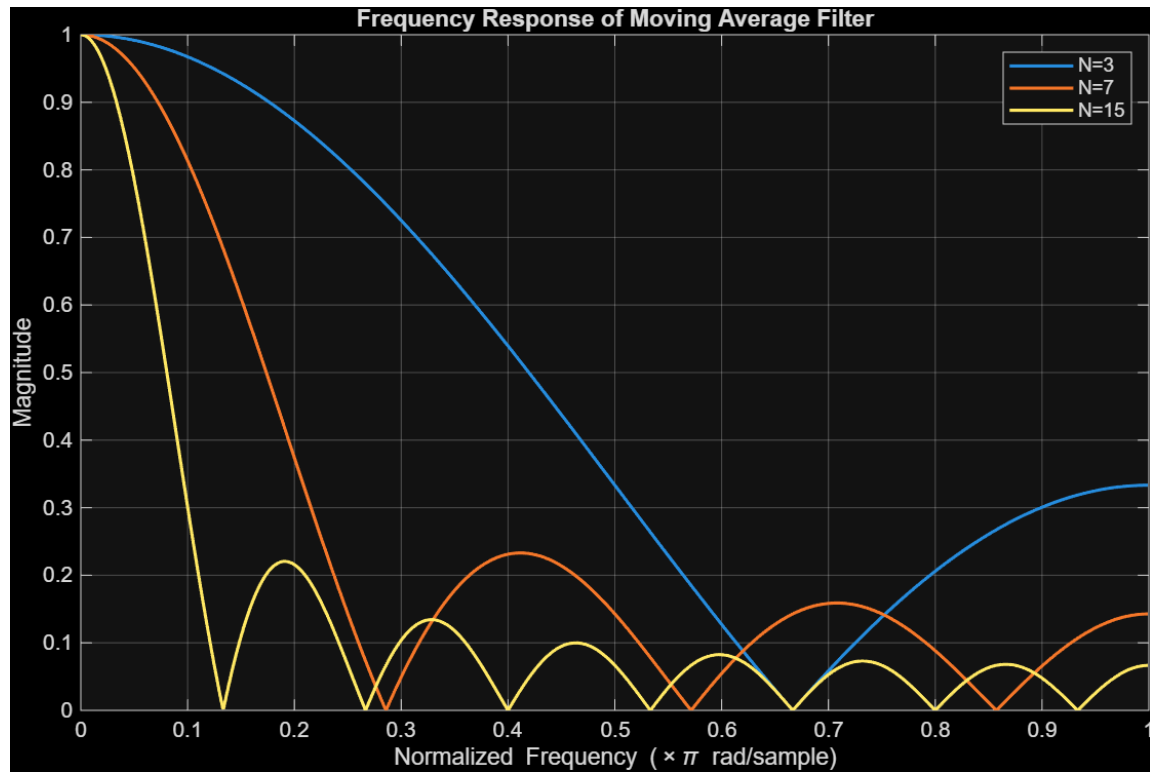
(b)

The zeros are the  $N$ -th roots of unity, located exactly on the unit circle at angles  $\theta_k = \frac{2\pi k}{N}$  for  $k = 1, 2, \dots, N-1$ . The zero at  $z = 1$  ( $k = 0$ ) is canceled by the system's single pole at the same location.



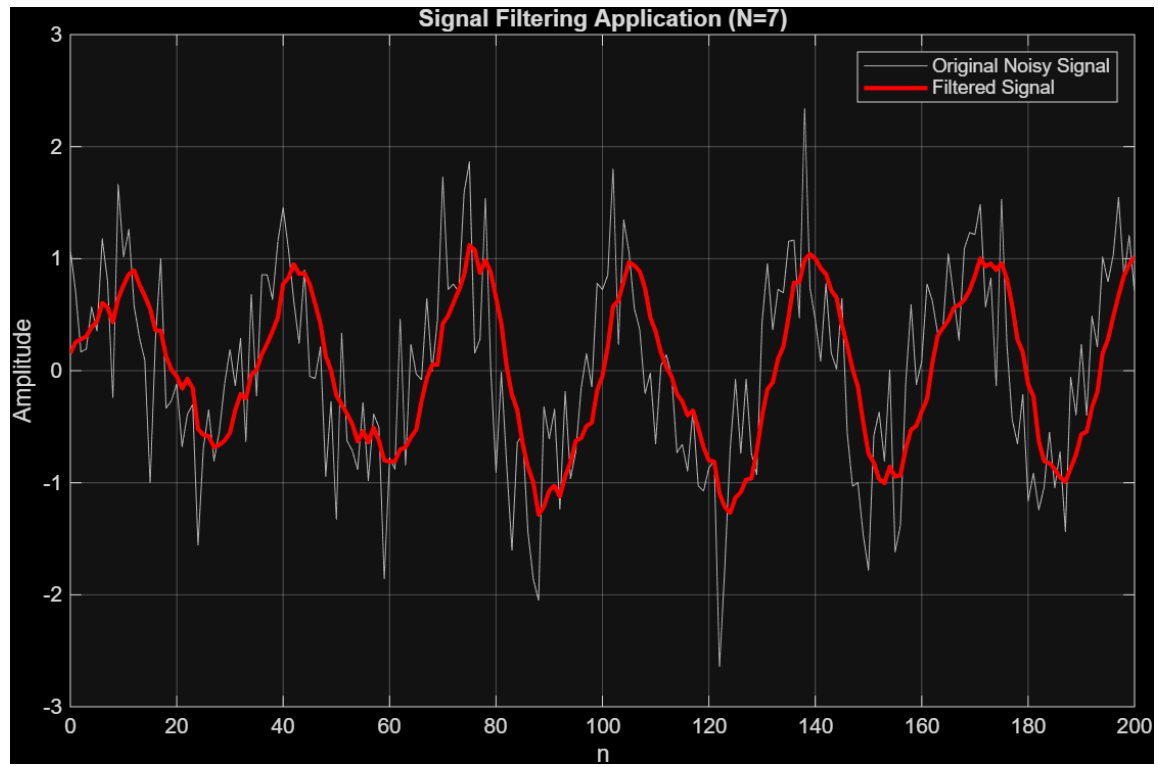
(c)

This acts as a **low-pass filter**. By **increasing**  $N$ , the main lobe of the frequency response becomes narrower. This effectively lowers the cutoff frequency and increases the attenuation of high-frequency noise components.





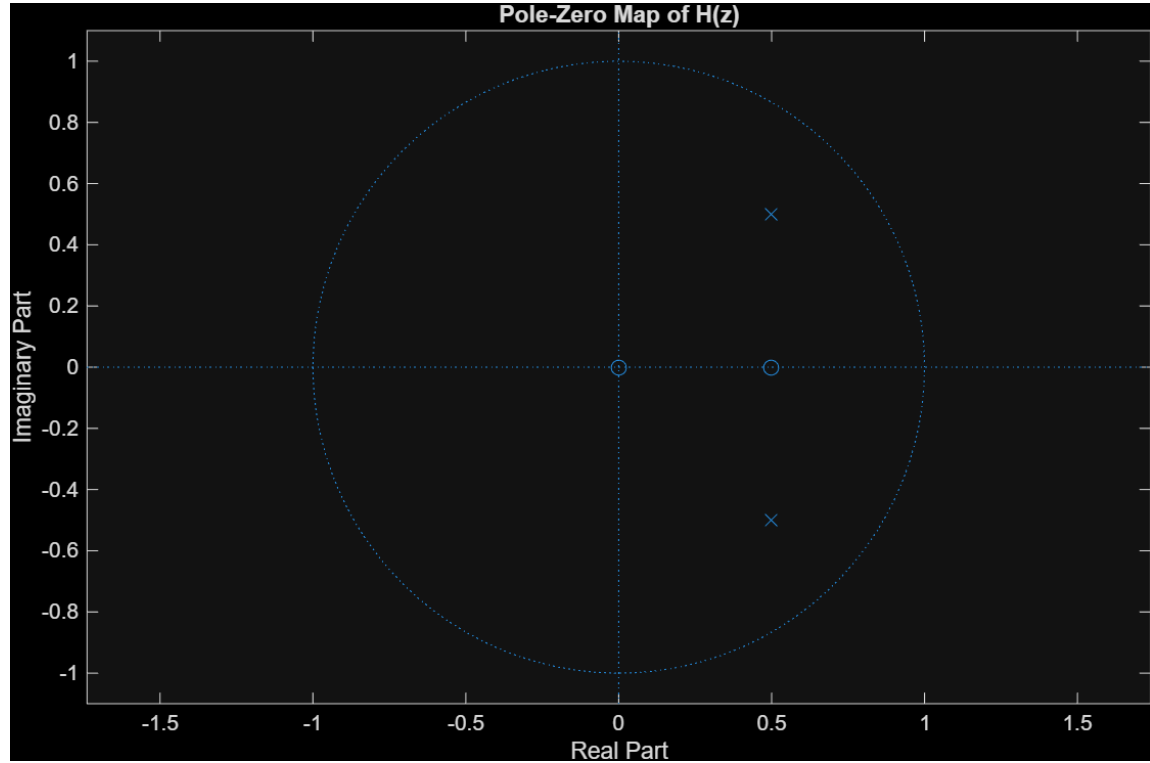
(d)



## 4. Z-Transform: Inverse via Partial Fractions

(a)

The transfer function is  $H(z) = \frac{1-0.5z^{-1}}{1-z^{-1}+0.5z^{-2}}$ . The roots of the denominator (poles) are located at  $p_{1,2} = 0.5 \pm 0.5j$ . The magnitude of these poles is  $|p| = \sqrt{0.5^2 + 0.5^2} \approx 0.707$ . Because the system is causal and the ROC ( $|z| > 0.707$ ) strictly includes the unit circle ( $|z| = 1$ ), the system is **BIBO stable**.



(b)&(c)

Using the residue coefficients obtained via partial fraction expansion, the complex poles can be converted into a real decaying cosine function. The closed-form expression for  $h[n]$  is:

$$h[n] = \sqrt{2} \left( \frac{1}{\sqrt{2}} \right)^n \cos \left( \frac{\pi}{4}(n-1) \right) u[n] \quad (3)$$

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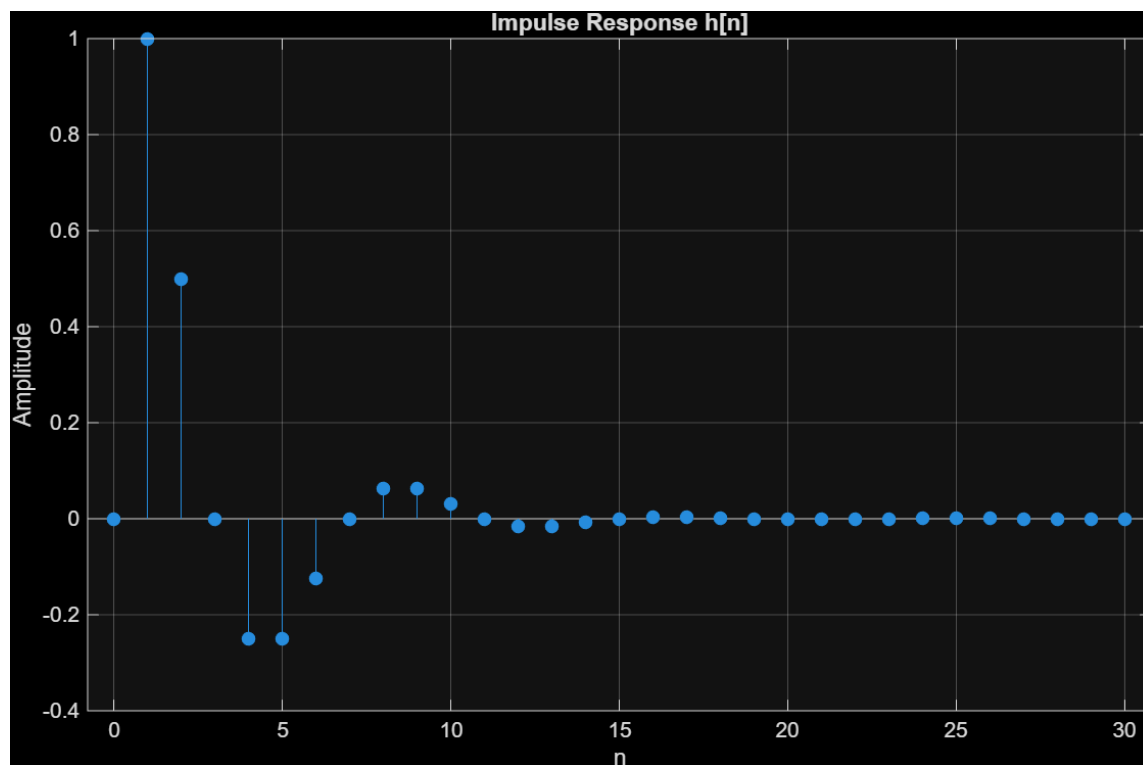
%part4_b
[r, p, k] = residuez(b, a);

%part4_c
syms z n
H_sym = (1 - 0.5*z^(-1)) / (1 - z^(-1) + 0.5*z^(-2));
h_n_sym = iztrans(H_sym, z, n);
disp('h[n] using iztrans:');
disp(h_n_sym);

```

(d)

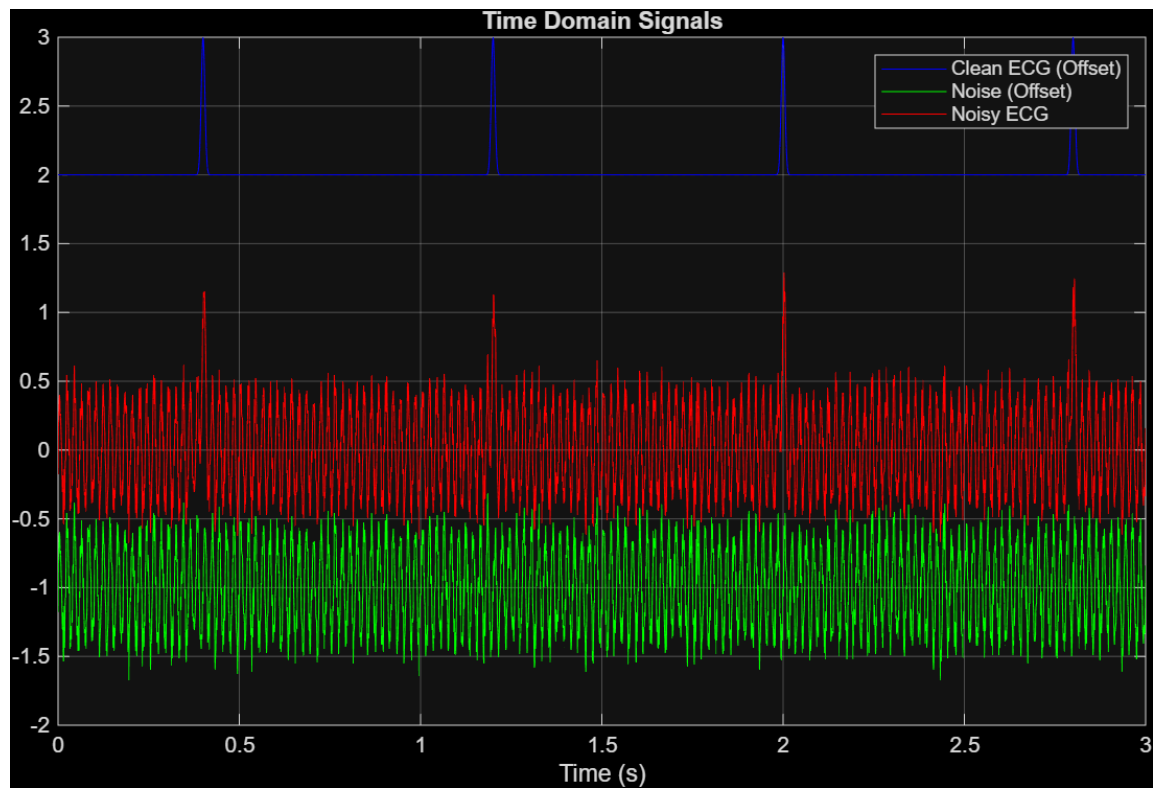
The plot of  $h[n]$  exhibits the shape of a **decaying sinusoid**. This transient behavior perfectly aligns with our stability analysis; since the amplitude of the oscillations decays to zero as  $n \rightarrow \infty$  ( $\sum |h[n]| < \infty$ ), the system is unequivocally stable.



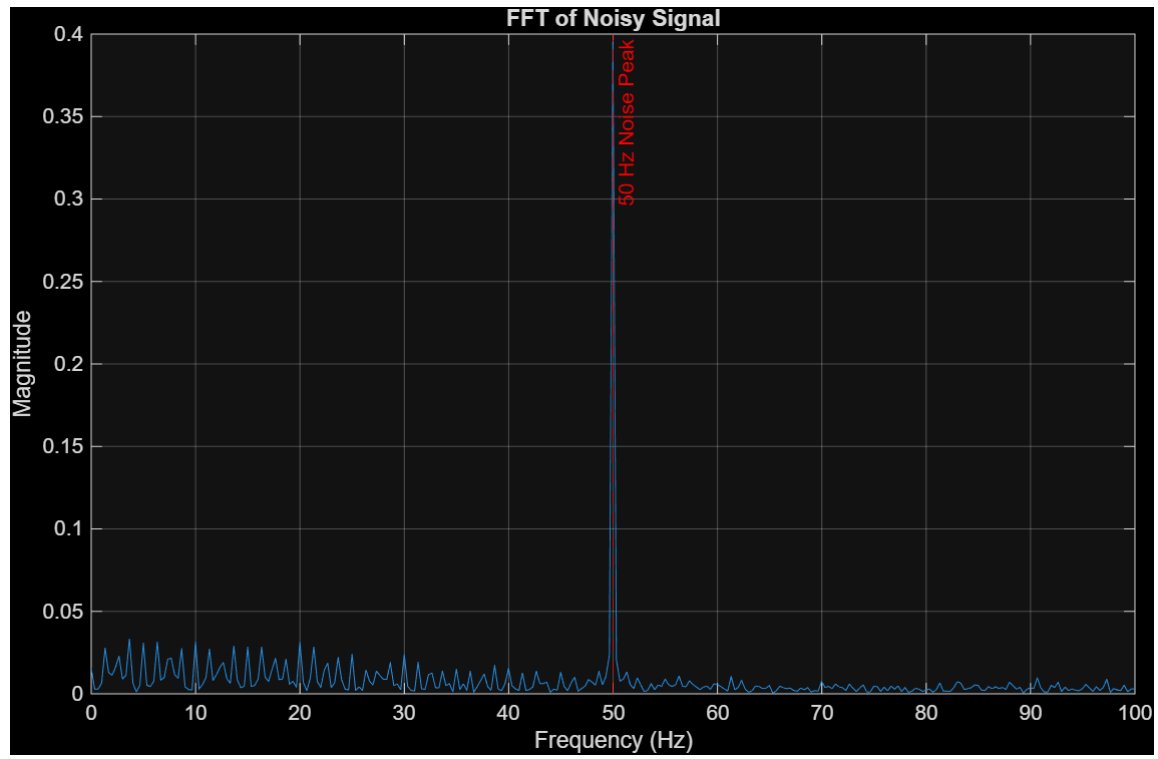
## 5. Application: ECG Denoising with a Notch Filter

(a)

Power-line interference is highly visible both in the time-domain signal and in its frequency spectrum. By applying the FFT to the synthetic noisy ECG, a prominent and isolated peak is distinctively visible exactly at **50 Hz**, which corresponds to the induced noise.



(b)

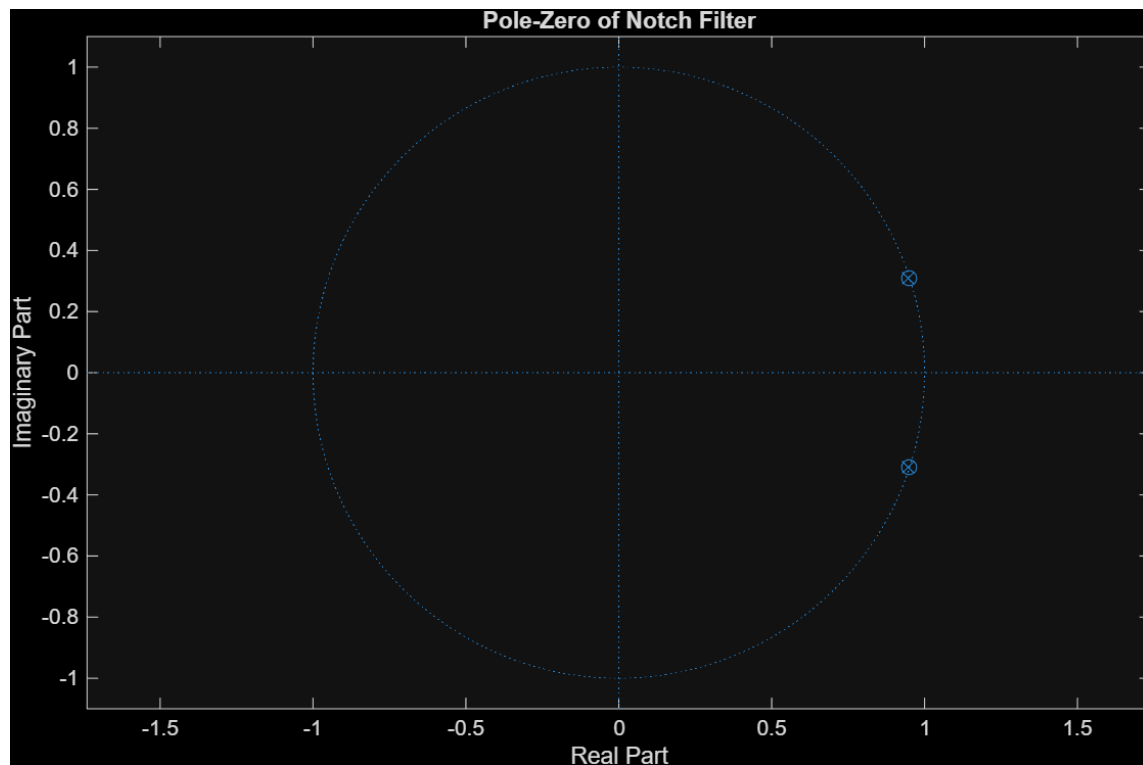


(c)

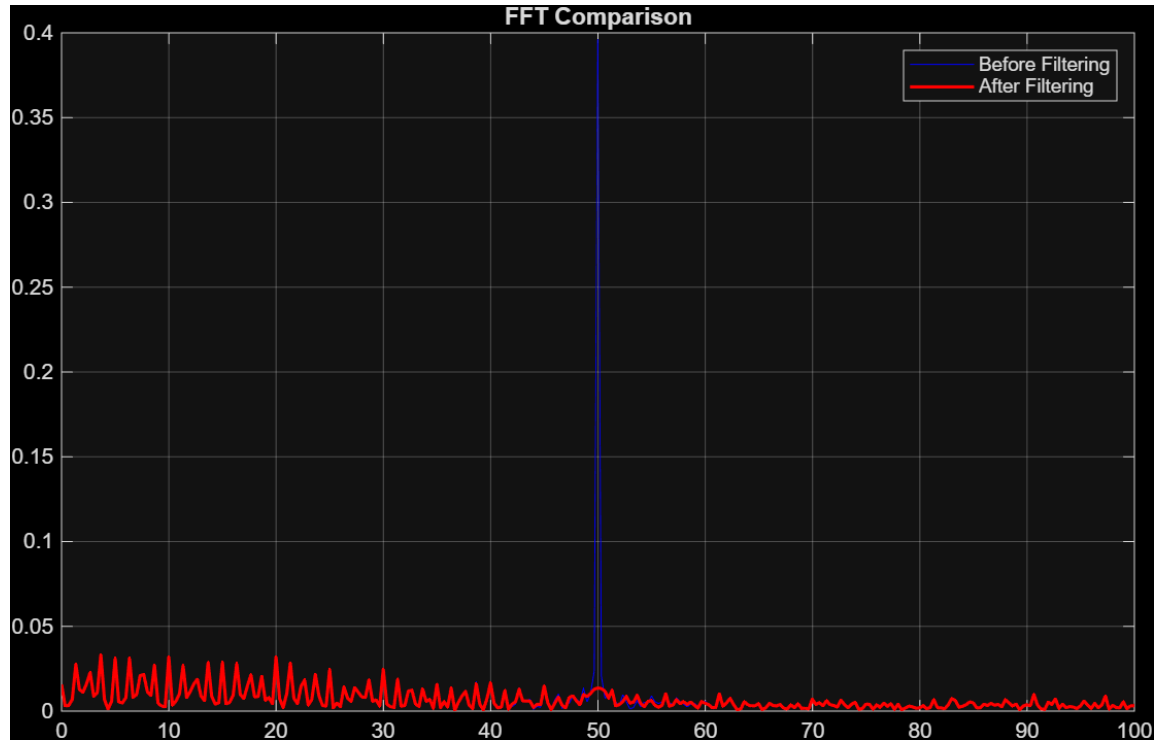
A digital notch filter was designed at  $f_0 = 50$  Hz.

**Zero Placement:** The zeros of the notch filter are placed exactly on the unit circle at the angular frequency corresponding to 50 Hz.

Because the distance from these zeros to the unit circle at that specific frequency is zero, the magnitude response perfectly drops to absolute zero, completely eradicating the 50 Hz interference.



(d)



(e)

The quality factor inversely dictates the rejection bandwidth of the notch filter.

- $Q = 5$ : The bandwidth is too wide, unnecessarily attenuating legitimate physiological ECG frequencies adjacent to 50 Hz.
- $Q = 100$ : The rejection band is extremely narrow, which is ideal for preserving the signal but requires high precision and makes the filter highly sensitive to slight frequency variations.
- $Q = 35$ : Provides an optimal engineering compromise between effectively rejecting the noise and preserving the clinical integrity of the ECG signal.

